

Phenol-Chloroform environmental DNA (eDNA) Extraction Protocol for *Okeanos Explorer* Specimens

Based on an original protocol by Meredith Everett (NOAA-NWFSC) with notes and modifications by Allen G. Collins (NOAA National Systematics Lab) and Steven R. Auscavitch (Smithsonian National Museum of Natural History).

Methodological Notes:

- Perform all steps (besides the centrifugations) in the fume hood with UV light in the Pre-PCR lab.
- Wear nitrile gloves at all steps. Wipe down with DNAaway frequently.
- Prior to starting work, wipe the work surface with a Hype-Wipe (contains bleach) and allow it to air dry. Next, wipe the surface with dilute ethanol (70%). Lastly, run the UV lamp in the hood for 15 minutes.
- Wipe pipettes with a DNAaway or a Hype-Wipe, followed by a wipe with a Kim-wipe or paper towel moistened with dilute ethanol. Place the pipettes in the fume hood prior to running the UV light.
- Use only pipette filter tips.
- Chill isopropanol and ethanol on ice prior to using.
- Allow PCI solution (Autogen Buffer R3) to equilibrate to room temperature for at least 30 min before using.

Materials needed:

- 200 µL and 1000 µL filtered tips (sterile)
- P200 and P1000 single-channel pipette
- 1.5 mL LoBind Eppendorf tubes (autoclaved, sterile)
- 5.0 mL LoBind Eppendorf tubes (autoclaved, sterile)
- Simport 2.0 mL Cryovial with Silicon washer (T310-2A)
- KimWipes

Reagents needed:

- DNAaway and/or Hype-wipes (with Bleach)
- Freshly made Longmire's buffer (for extraction blank)
- Proteinase K solution (20 mg/ml) (e.g. Qiagen Proteinase K or Autogen supply)
- Autogen Buffer R3 (Phenol chloroform isoamyl alcohol)
- Isopropanol (100%)
- Ethanol (70%)
- Autogen resuspension buffer (R9)

Procedure

Digestion:

1. Turn on the VorTemp incubator and set it to 56°C with a 50 RPM agitation speed.

2. Remove 5 mL eDNA preservation tubes containing 0.45 micron nitrocellulose filters and Longmire's lysis buffer (2-3 ml) from the -80 freezer. Thaw to room temperature.
3. Wipe tubes down with a Kimwipe moistened with DNAaway or a hypewipe, followed by a moistened Kimwipe with 70% ethanol. Ensure that sample labels remain visible.
4. Add 150 μ L of Proteinase K to each 5 ml tube to a final concentration of ~1mg/ml to each sample. Invert tube 5 times to mix. Ensure the filter remains immersed in the lysis buffer.
5. Incubate the samples at 56°C overnight (20-24 hours) with agitation. Record the length of the incubation.

Extraction:

6. Remote the samples from the VorTemp and shake vigorously by hand 5 times.
7. Spin down the samples briefly at 13,300 g in the chilled centrifuge set to 4°C.
8. Recover 2.5 ml of the digestion mixture (without the filter) and transfer to a second labeled 5 mL sterile LoBind tube.
 1. Reserve 0.5-1 ml (depending on starting volume) of digestion lysate and filter for the biorepository archive voucher. Refrigerate until ready.
9. Add 2.5 ml of PCI (Buffer R3) to the digestion mixture.
10. Invert five times to mix thoroughly.
11. Spin at 13,300 g for 5 min in a chilled centrifuge set to 4°C.
12. Transfer 4 mL of the aqueous layer into two separate 5.0 mL LoBind tubes (2 mL each) for isopropanol precipitation.
13. Add 1.2 mL (0.6X) of chilled 100% isopropanol to each tube of recovered aqueous layer and invert to mix.
14. Incubate in the fridge at 4°C for at least 2 hours.
15. Spin isopropanol precipitation at 13,300 g for 30 min in a chilled centrifuge set to 4°C.
16. Carefully decant isopropanol.
17. Add 1,000 μ L chilled 70% EtOH.
18. Spin at 13,300 g for 5 min in a chilled centrifuge set to 4°C.
19. Carefully decant off most of the ethanol.
20. Repeat step 18 for a second ethanol wash.
21. Briefly spin down the tubes to collect the sample at the bottom of the tube.
22. Carefully pipette off any remaining ethanol without disturbing the pellet (Caution: pellet may not be visible).
23. Allow the pellet to air dry in the hood for 30 minutes.
24. Resuspend the DNA in 25 μ L of Autogen Resuspension buffer (R9).
25. Spin the samples and allow the DNA to rehydrate at room temperature for ~1 hour or overnight at 4°C in the fridge.
26. Combine the 25 μ L volumes into a single 50 μ L extract in a new 1.5 mL Eppendorf tube labeled with the sample number, date, and your initials.
27. OPTIONAL: Aliquot a working volume of 10-20 μ L for Qubit quantification (use 2 μ L) and PCR into the second labeled 1.5 mL Eppendorf tube.

NMNH Biorepository archiving:

28. Sterilize a pair of forceps using DNAaway followed by dilute ethanol (70%). Dry under a UV lamp for at least 15 minutes or until ready to use. Re-sterilize forceps as necessary between samples.
29. Using sterile forceps remove the filter from the 5.0 mL LoBind sample tube and transfer to a Simport cryovial.
30. Transfer the remaining lysate using a P1000 pipette. If necessary, spin down the tube to concentrate lysate to the bottom of the tube.
31. Assigned a biorepository number to each sample and affix the corresponding barcode sticker, recording it in
32. Store cryovials at -80°C until transfer to the Biorepository at MSC.